Privacy-Preserving Human Activity Recognition from mmWave Radar Point Clouds using Lightweight Point Transformers

Team 10 | 23I006 Deep Learning Mini Project

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Why radar for activity recognition?

Camera

Accurate, but records identifiable images; fails in darkness; low acceptance at home

Wearables

No images, but must be worn and charged; often forgotten by elderly users

mmWave radar

Contact-free, works in the dark, no image formed: sparse 3-D points + velocity

- Goal: recognise activities from radar point clouds alone, accurately and cheaply enough for an edge device
- Applications: elderly activity monitoring, smart homes, privacy-preserving indoor sensing

Problem statement

Sparse

median 21 (RadHAR) and 6 (MiliPoint) returns per frame

Noisy

34.6% antenna echoes; 50.1% of the rest is static clutter

Temporal

the activity lives in the motion across frames, not in one frame

Leaky evaluation

overlapping windows put near-duplicates in train and test

Motivation: the same dataset, 34 % or 98 %?

34.45 %

MiliPoint authors, PointNet++ (NeurIPS 2023)

98.25 %

MAK-GCN 2025, random 80/10/10 split

72.90 %

of test frames already seen in training under random window splits (our audit)

- Need 1: a lightweight, noise-robust model that exploits radar structure
- Need 2: leakage-free evaluation on unseen recordings and unseen people

Existing approaches

Work	Method	Reported accuracy	Note
RadHAR 2019	Voxels + TD-CNN + BiLSTM	90.47 % (RadHAR)	dataset paper
Lee & Kim 2022	GNN + pre-trained 3-D joints	96.55 % (RadHAR)	needs joint labels
Gu et al. 2024	Lite-PointNet + BiLiLSTM	95.12 % (RadHAR)	79.7 K params
MAK-GCN 2025	Adaptive graph convolution	97.54 % RadHAR / 98.25 % MiliPoint	random split on MiliPoint
OG-PCL 2025	Occupancy-gated convolution	91.75 % (RadHAR)	0.83 M params, 5-fold CV
MiliPoint 2023	PointNet++, DGCNN, PointMLP	34.45 % (49 actions)	dataset paper

Research gap and objectives

Research gap

- Dense-point backbones or voxels ignore radar structure (sparsity, Doppler, clutter)
- Inconsistent, often leaky protocols
- Robustness, transfer and edge cost rarely reported together

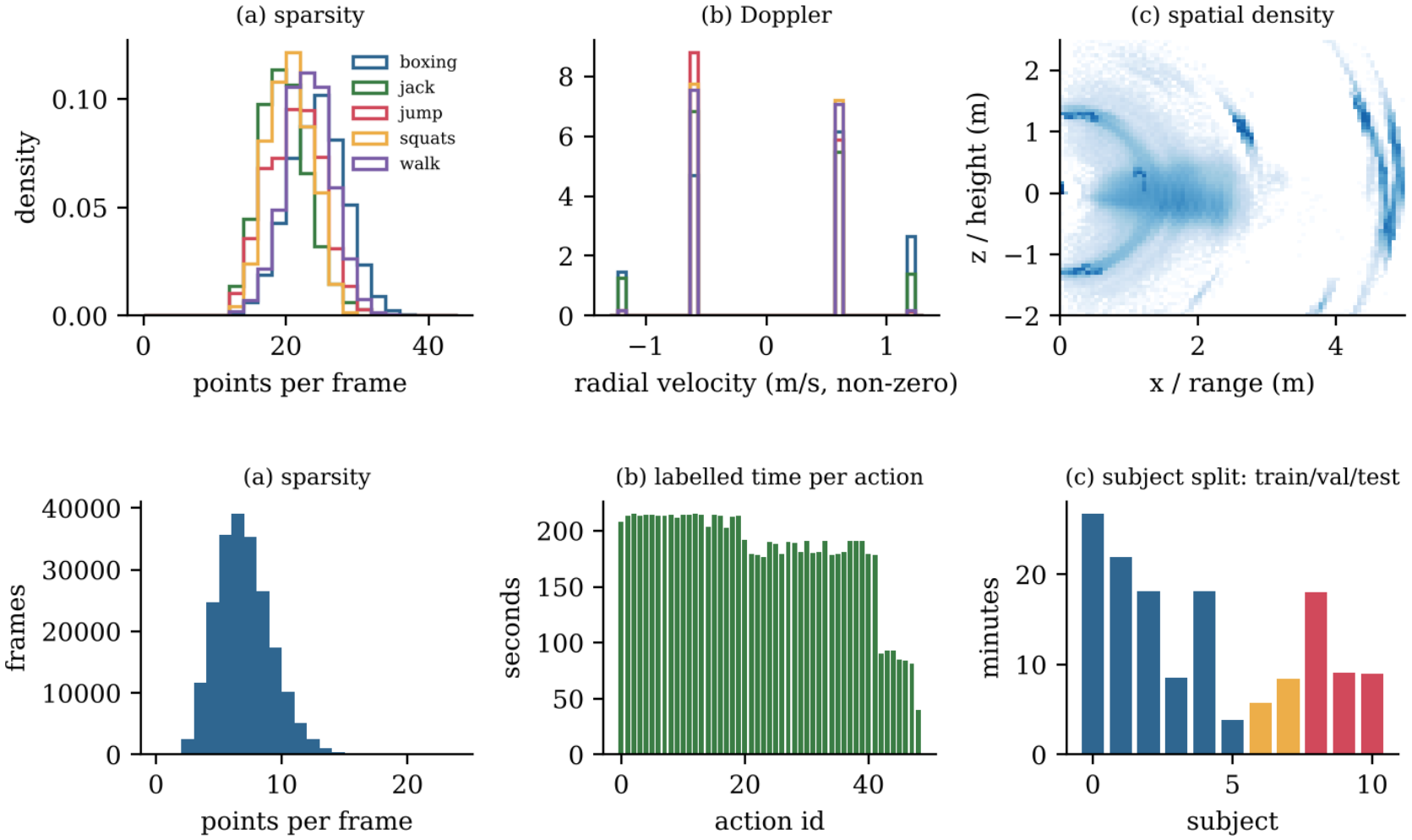
Objectives

1. Lightweight spatio-temporal point transformer
2. Radar-specific clutter removal
3. Leakage-free protocols + automated audit
4. Seeds, significance, ablation, robustness, transfer, XAI
5. Saved model + working demonstration

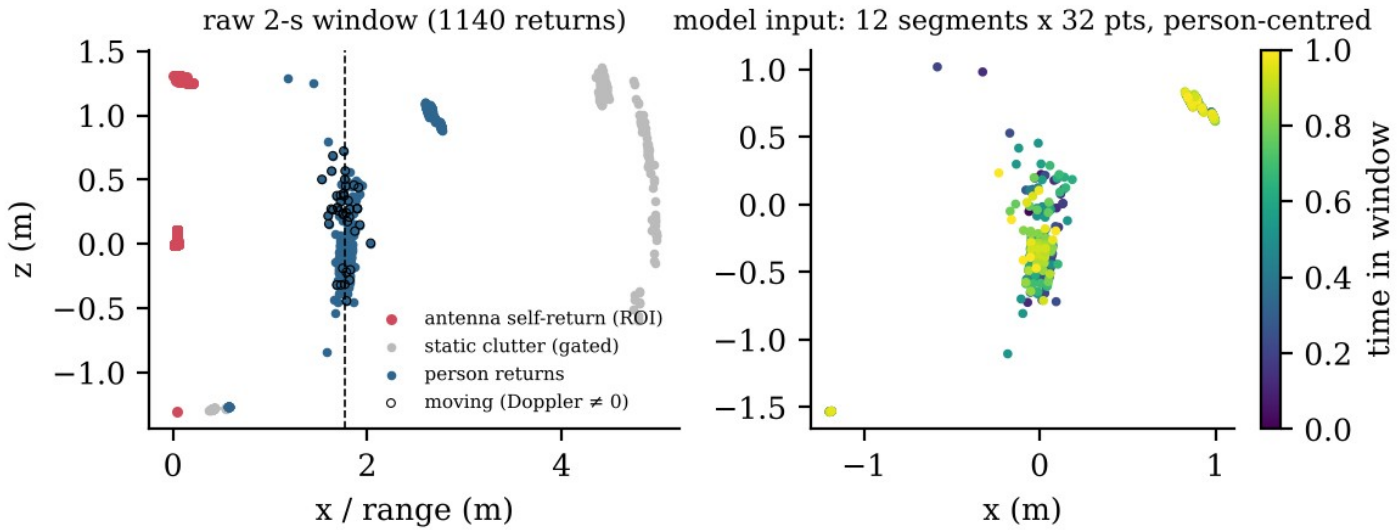
Datasets

	MiliPoint (primary)	RadHAR (secondary)
Radar / rate	TI IWR1843, 24 Hz	TI IWR1443, 30 Hz
Subjects	11 (IDs available)	2 (no IDs)
Classes	49 actions	5: boxing, jumping jacks, jump, squats, walk
Size	545,059 frames; 212,920 labelled	200 recordings; 71.5 + 21.4 min
Points / frame	median 6	median 21
Features	x, y, z	x, y, z, Doppler, intensity
Our test set	subjects 8–10 (unseen people)	official test recordings

Dataset challenges (measured)



Radar-specific pre-processing



- ROI: remove antenna echoes (x < 0.3 m)
- Centre on moving (Doppler ≠ 0) returns
- Gate: keep ≤ 1.2 m from the person
- 2-s window → 12 segments × 32 points
- Train-only intensity statistics

- Augmentation (training only): rotation ±15°, scaling 0.9–1.1, 1 cm jitter, re-sampling; optional noise-robust mode: density, dropout, false detections, segment loss

Noise: measured vs. controlled

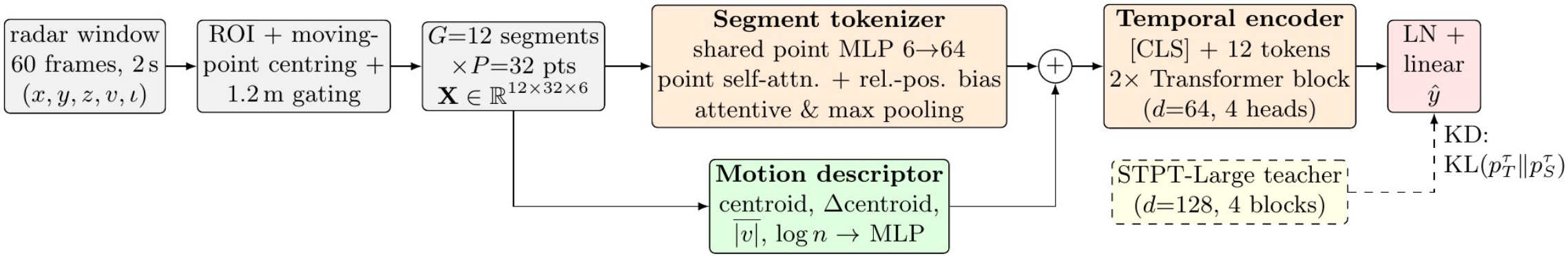
Measured in the data	Handled by
Antenna self-returns (34.6 %)	ROI filter
Static clutter / multipath	Doppler centring + gating
Sparse, variable density	4–5-frame segments + re-sampling
Quantised Doppler	kept as a feature

Injected into the test set	Levels
Coordinate jitter	σ = 2, 5, 10 cm
Point dropout	5, 15, 30 %
Increased sparsity	keep 50, 25, 12.5 %
Outlier injection	5, 10, 20 %
Variable density	min keep 0.5, 0.25, 0.1
Frame loss	25, 50 % of segments
Combined	severity 1–3

Data-leakage prevention and audit

Check	Result	Status
RadHAR recordings shared between splits	0 / 0 / 0	PASS
Identical frames across splits	0	PASS
Train-only normalisation	verified	PASS
RadHAR session independence	**median gap 29 s, same days**	**NOT INDEPENDENT**
MiliPoint subjects shared (subject split)	none	PASS
MiliPoint random split: test frames seen	**72.90 %**	**LEAKY**
Checkpoint = best validation epoch	0 / 0 runs	PASS

Proposed model: STPT-Lite



0.0 K
parameters

n/a
MFLOPs per 2-s window

n/a ms
CPU latency, 1 thread, batch 1

n/a MB
INT8 weights

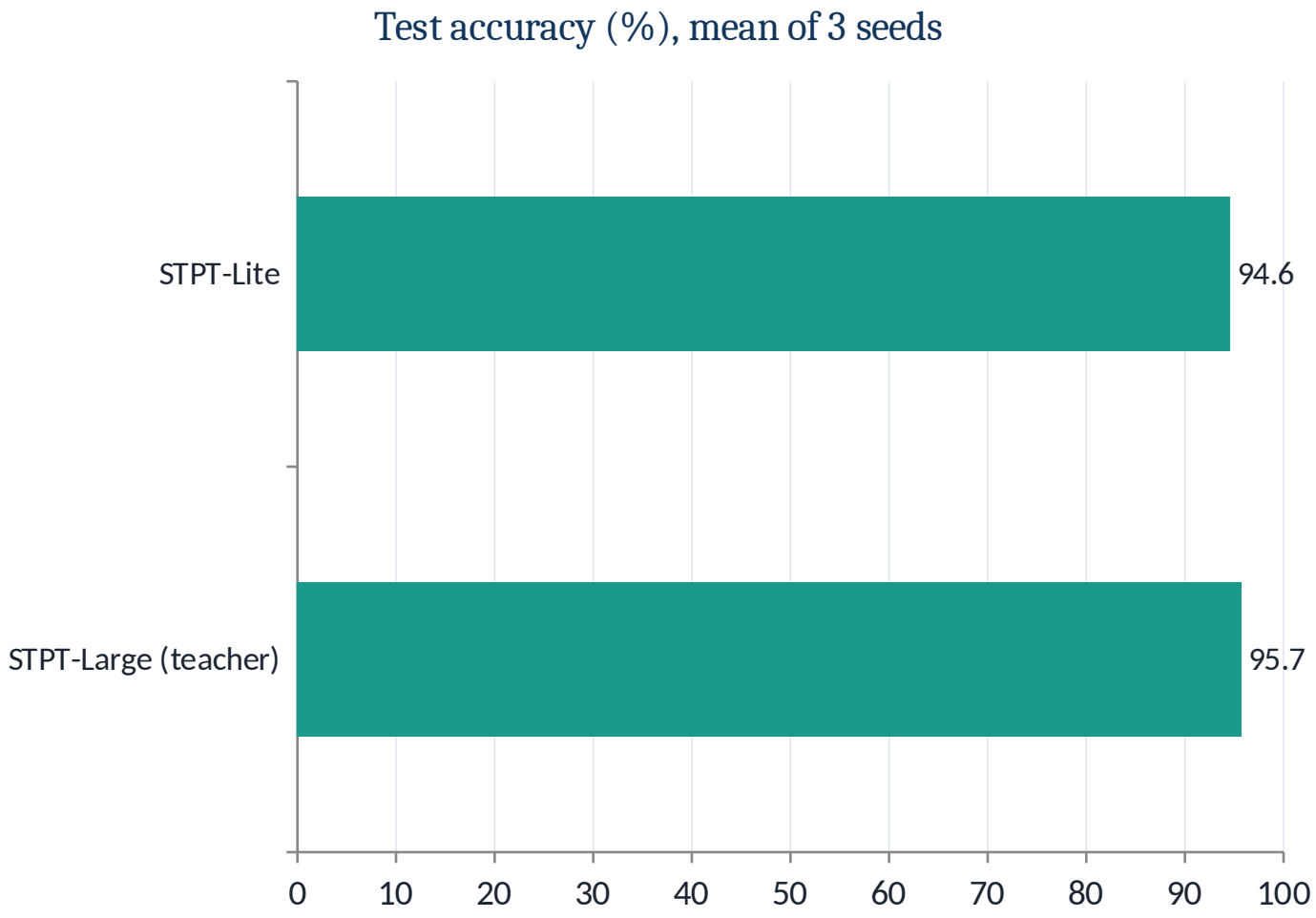
Model selection: baselines under one protocol

Model	Idea	Params (K)
STPT-Large (teacher)	factorised attention, d = 128	725.3
STPT-Lite	factorised attention, d = 64	117.5

Training methodology and overfitting control

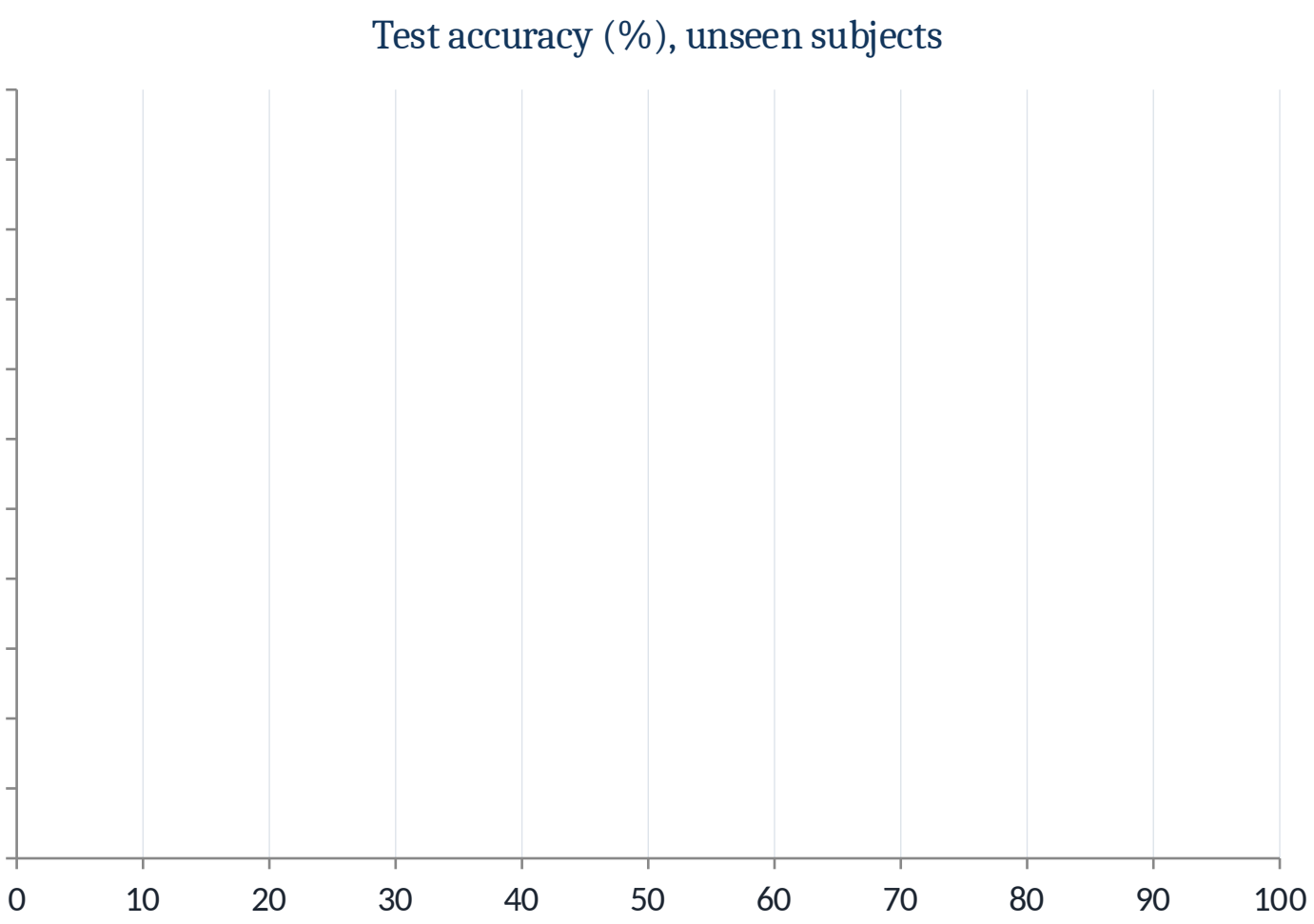
- AdamW, one-cycle LR (peak $2 \cdot 10^{-3}$), weight decay 0.05
- Label smoothing 0.1, gradient clipping 1.0
- ≤ 15 epochs, early stopping (patience 5) on val macro-F1
- Best-validation checkpoint; test evaluated once
- 3 seeds; mean ± std reported
- KD: $\alpha = 0.7$, $T = 4$

RadHAR: n/a % with 0 K parameters



Model	Acc. (%)	F1 (%)
STPT-Large (teacher)	95.74 ± 1.07	95.78

MiliPoint, unseen people: n/a % on 49 actions



Model	Top-1 (%)	Top-3 (%)

Statistical validation and error analysis

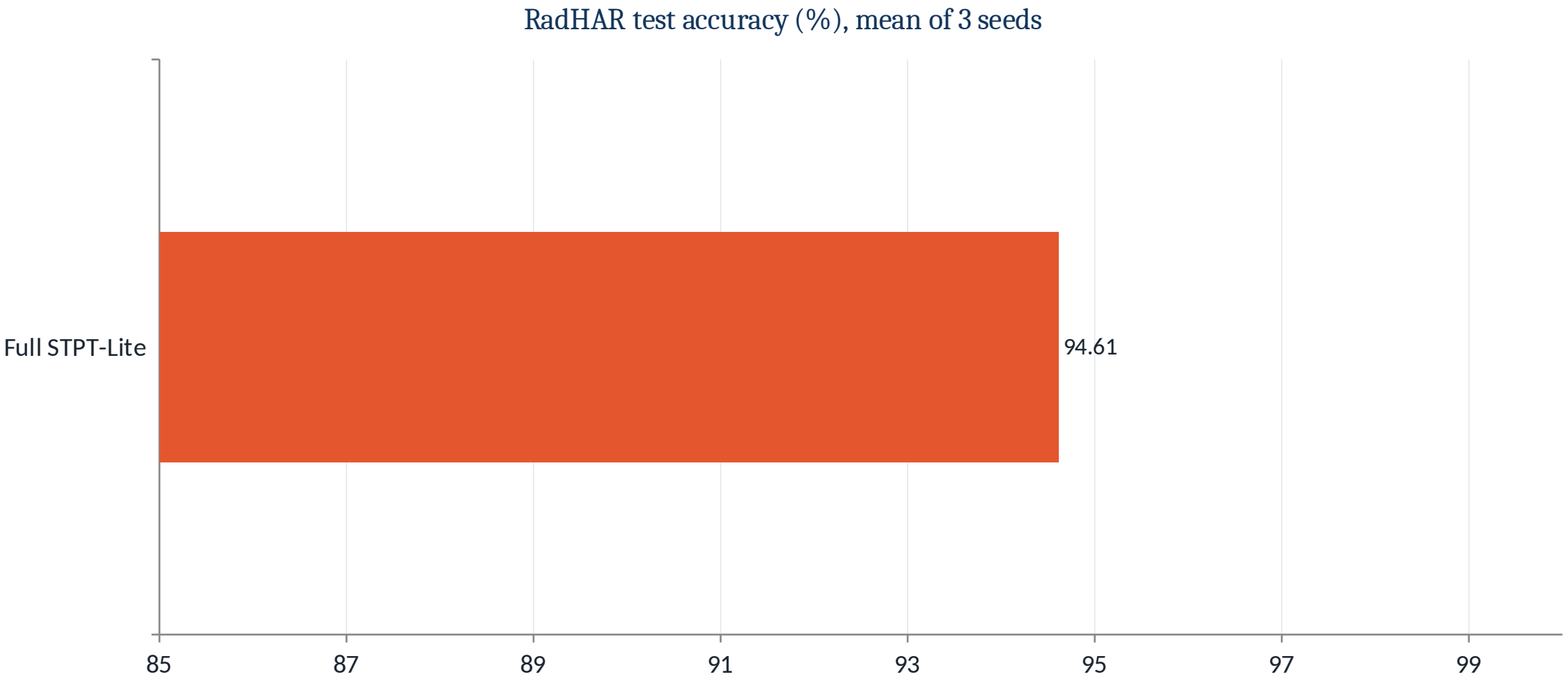
vs. (RadHAR, seed 0)	Acc.	p (Holm)

Noise robustness

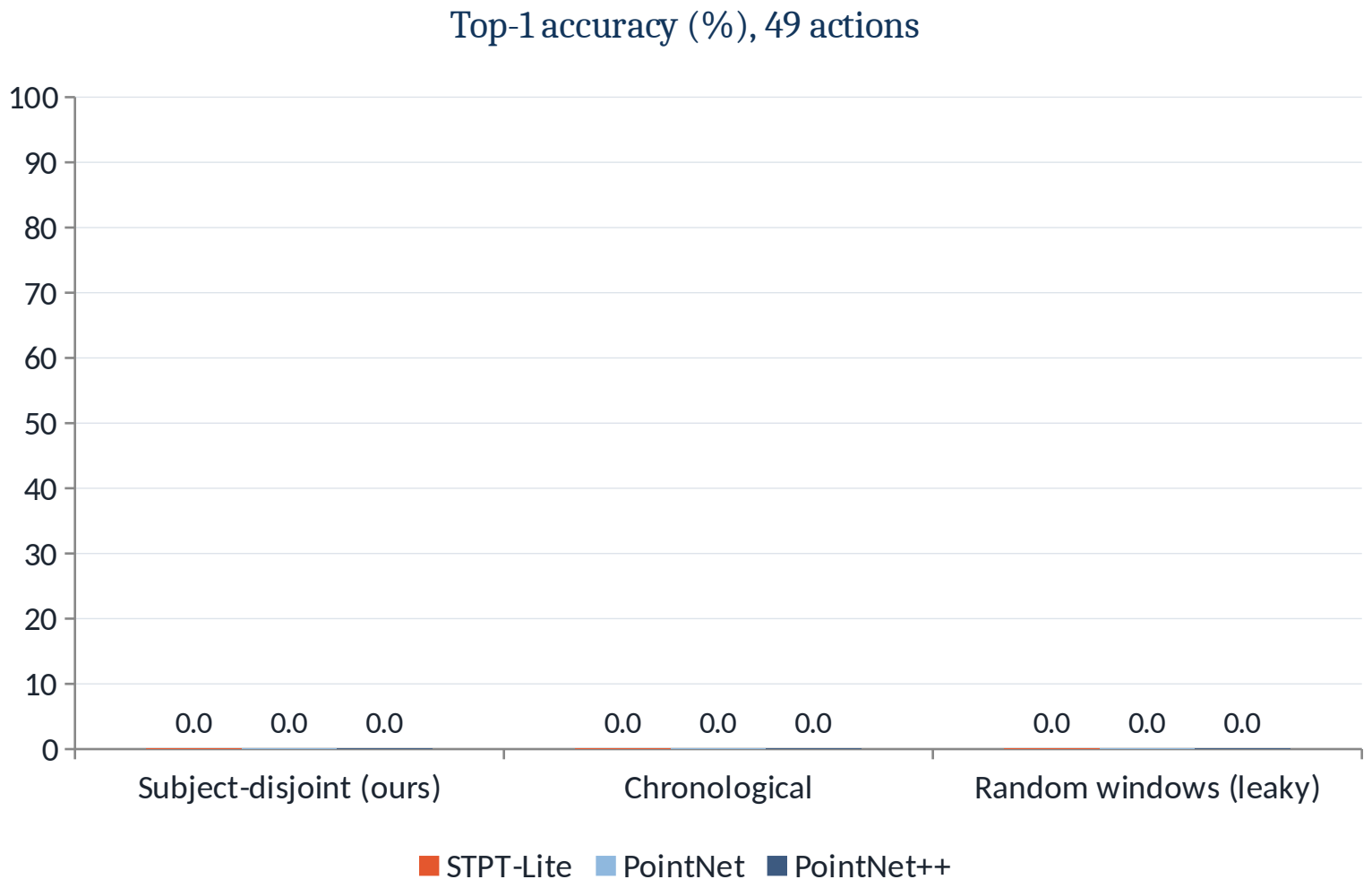
Computational efficiency

	Params (K)	MFLOPs	Latency (ms)	FP32 (MB)	INT8 (MB)	INT8 acc. (%)
stpt_lite_kd	**0.0**	**n/a**	**n/a**	**n/a**	**n/a**	**n/a**
stpt_tiny_kd	0.0	n/a	n/a	n/a	n/a	n/a
n/a	0.0	n/a	n/a	n/a	n/a	n/a

Ablation study (RadHAR, 3 seeds)



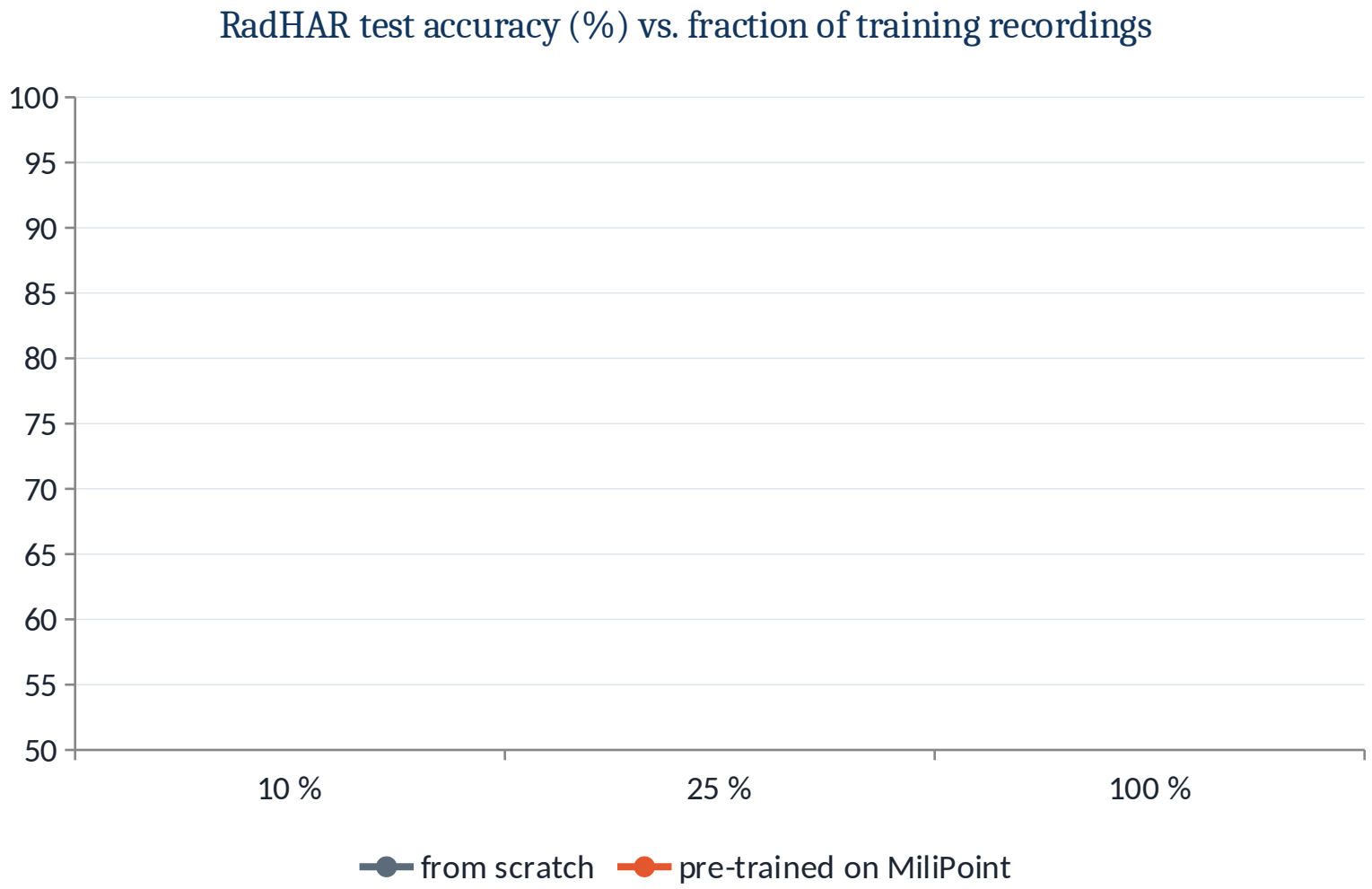
How the split changes the answer (MiliPoint)



Takeaway

Random window splits leak 72.90 % of test frames into training. Report subject-disjoint accuracy for claims about new users.

Cross-dataset transfer: MiliPoint → RadHAR

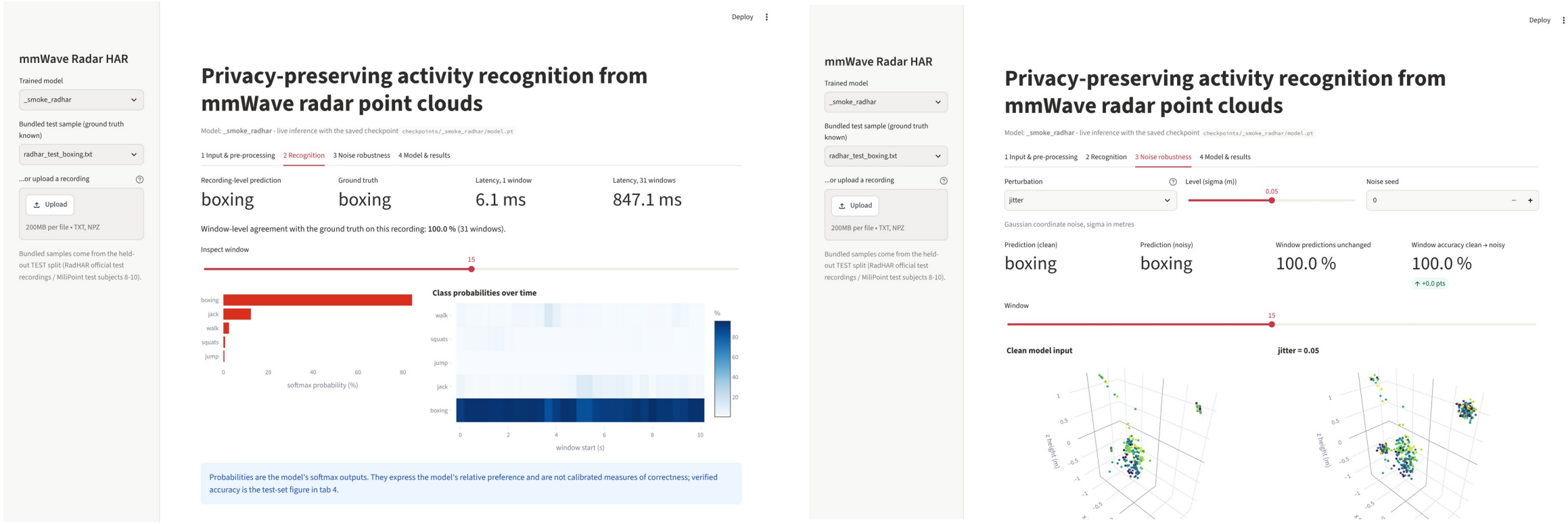


Reading the plot

Gap between the lines = value of pre-training. It is largest when little target data is available, if transfer helps.

Explainability

Working demonstration (Streamlit)



- Upload raw radar file → pre-processing view → live prediction → noise injection

Limitations and future work

Limitations

- RadHAR: same people and sessions in test
- MiliPoint: xyz only; 3 test subjects
- Synthetic noise, not physical simulation
- Desktop-CPU latency; no MCU deployment
- No fall labels; single person

Future work

- Leave-one-subject-out, cross-room data
- Deploy on the radar companion processor
- Physics-based radar noise simulation
- Self-supervised pre-training, streaming recognition

Conclusion

- STPT-Lite+KD: n/a % on RadHAR and n/a % on 49 MiliPoint actions of unseen people
- 0.0 K parameters, n/a MFLOPs, n/a ms, n/a MB INT8
- Doppler-based clutter removal; automated leakage audit; quantified protocol inflation
- Saved model, working demo, paper, technical report and full logs

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